Visiting Paris

A Multimodal Optimization Framework for Travel Itinerary Planning

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Executive Summary

Urban tourism itinerary planning in large metropolitan destinations such as Paris remains highly inefficient due to fragmented information, heterogeneous opening schedules, transportation trade-offs, and limited traveler resources (budget and time).

This project proposes a Multimodal Orienteering Problem with Time Windows (OPTW), formulated as a Mixed-Integer Linear Programming (MILP) model, that simultaneously optimizes tourist preferences, time constraints, budget limitations, and transportation modality choices.

Unlike traditional recommendation platforms, the proposed framework generates an operationally feasible and mathematically optimized itinerary that maximizes user satisfaction under realistic urban conditions.

The framework incorporates a dynamic transportation rule: whenever walking time between attractions exceeds 30 minutes, metro transportation is prioritized in order to improve route efficiency and reduce travel fatigue.

The proposed solution is computationally viable, scalable, and adaptable to broader smart tourism applications.

1. Problem Context

Paris is one of the most visited cities in the world, renowned for its vast number of tourist attractions and the considerable distances between many of them. Large metropolitan destinations such as Paris present significant logistical challenges for tourists due to the high concentration of attractions, heterogeneous opening hours, transportation complexity, and travelers' limited time and budget resources.

Most tourists face three primary constraints:

- Limited available time
- Restricted travel budgets
- Information overload during itinerary planning

As a result, visitors frequently rely on heuristic decision-making through fragmented platforms such as maps, blogs, and tourism websites.

This often leads to:

- Excessive travel time between attractions
- Poor sequencing of visits
- Underutilization of attraction opening windows

- Reduced overall travel satisfaction

These inefficiencies become more pronounced in dense urban tourism ecosystems, where mobility decisions strongly affect itinerary feasibility and quality.

2. Gaps in Current Solutions

Existing tourism platforms primarily focus on recommendation rather than optimization.

Platform Type	**Main Limitation**
Navigation applications	Optimize routes but not overall travel satisfaction
Recommendation platforms	Suggest attractions without itinerary optimization
Static itineraries	Lack personalization and adaptability
AI travel assistants	Provide qualitative guidance without mathematical optimization

Table 1: Limitations of Existing Tourism Planning Platforms

No existing solution simultaneously manages:

- User preferences
- Time windows
- Budget constraints
- Transportation mode selection
- Spatial efficiency

These gaps suggest that existing platforms tend to focus on only one of the limitations and constraints faced by users. However, they do not propose an integrated algorithm capable of addressing all these constraints within a single application.

3. Methodology

The proposed framework models itinerary planning as a constrained optimization problem applying **Orienteering Problem with Time Windows**. The model receives the following user inputs:

- Ranked tourist attraction preferences
- Daily budget
- Available travel time

Using these inputs, the system generates an itinerary that maximizes tourist utility based in the ranking that the user assign to each attraction while satisfying operational constraints.

The optimization incorporates three main dimensions:

3.1. Temporal Feasibility

Considering that each attraction has different opening and closing hours, the proposed algorithm explicitly incorporates these temporal constraints into the optimization process. Therefore, attractions may only be visited within their respective operating hours.

3.2. Spatial Efficiency

The model explicitly considers travel times between attractions during route construction in order to ensure temporal feasibility and improve overall itinerary efficiency. This allows the algorithm to minimize unnecessary displacement, optimize attraction sequencing, and reduce total travel fatigue for tourists.

3.3. Dynamic Transportation Selection

The model dynamically selects between walking and metro transportation in order to balance efficiency, accessibility, and travel time optimization. If the walking time between two attractions exceeds 30 minutes, metro transportation is automatically prioritized to reduce travel fatigue, improve route efficiency, and ensure compliance with overall itinerary time constraints.

The following figure illustrates the overall functioning of the algorithm and is intended to support the understanding of the next sections, where the different steps of the method are discussed in detail.

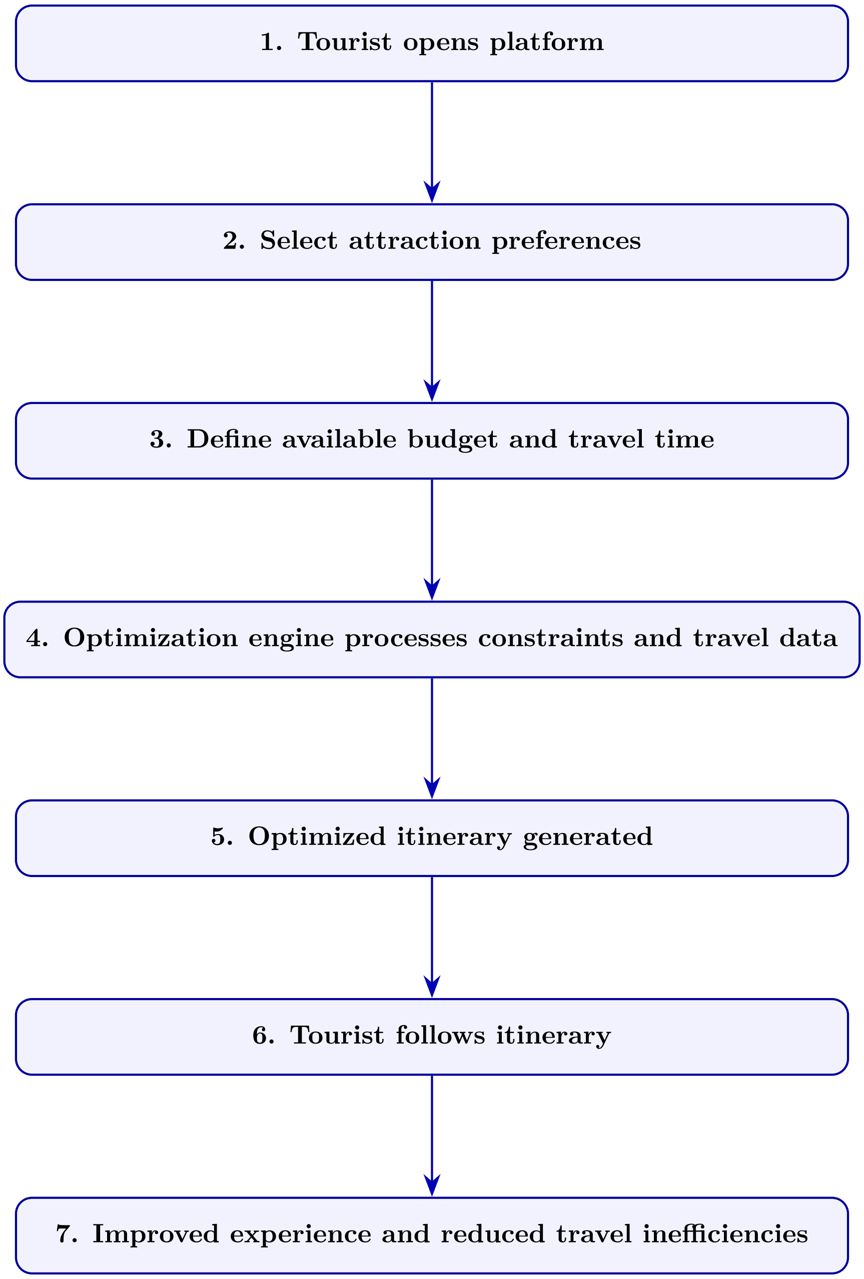

Figure 1: User Journey for the Tourism Optimization Framework

4. Key Contributions

The proposed framework contributes to urban tourism optimization in four main dimensions:

- Integration of user preferences, time constraints, budget limitations, and spatial structure into a unified optimization model
- Explicit incorporation of operational constraints to ensure itinerary feasibility
- Multimodal transportation modeling through walking and metro integration
- Transferability to other metropolitan tourism environments

5. Mathematical Formulation

5.1. Sets and Indices

- $I = \{0, 1, \ldots, 10\}$: set of nodes, where node 0 represents the hotel and node 10 the return node
- $V = \{1, \ldots, 9\}$: set of tourist attractions

5.2. Parameters

Parameter	Description
p_i	Cultural value of site i
s_i	Visit duration at site i
a_i, b_i	Opening and closing times
tw_{ij}	Walking travel time between nodes
tm_{ij}	Metro travel time between nodes
T^{max}	Maximum tour duration
C^{max}	Budget limit
$Ticket$	Metro fare cost
$Ticket\_M_j$	Attraction entrance cost
M_{cost}	Big-M constant for budget constraints
M_{time}	Time Big-M Constant

Table 2: Model Parameters

To avoid numerical instability and improve computational efficiency within the GLPK solver, the Big-M parameters M_{cost} and M_{time} are not assigned arbitrarily large values. Instead, they are dynamically computed as tight upper bounds of the system by combining the maximum potential journey durations and financial limits of the network, thereby strengthening the linear relaxation.

5.3. Decision Variables

- $x_{ij} \in \{0, 1\}$: walking arc between nodes i and j
- $y_{ij} \in \{0, 1\}$: metro arc between nodes i and j
- $u_i \geq 0$: arrival time at node i
- $c_i \geq 0$: cumulative transportation cost at node i

5.4. Objective Function

$$\max Z = \sum_{i \in V} p_i \left(\sum_{\substack{j \in I \\ j \neq i}} (x_{ij} + y_{ij}) \right) \tag{1}$$

5.4.1. Route Structure

$$\sum_{\substack{j \in I \\ j \neq 0}} (x_{0j} + y_{0j}) = 1 \tag{2}$$

$$\sum_{\substack{i \in I \\ i \neq 10}} (x_{i10} + y_{i10}) = 1 \tag{3}$$

$$x_{10} + y_{10} = 0 \tag{4}$$

$$\sum_{\substack{i \in I \\ i \neq k}} (x_{ik} + y_{ik}) = \sum_{\substack{j \in I \\ j \neq k}} (x_{kj} + y_{kj}) \quad \forall k \in V \tag{5}$$

$$\sum_{\substack{i \in I \\ i \neq k}} (x_{ik} + y_{ik}) \leq 1 \quad \forall k \in V \tag{6}$$

$$x_{ij} + y_{ij} \leq 1 \quad \forall i \neq j \tag{7}$$

5.4.2. Time Constraints

$$u_0 = 0 \tag{8}$$

$$\begin{gathered} u_j \geq u_i + s_i + tw_{ij} \cdot x_{ij} + tm_{ij} \cdot y_{ij} - M_{\text{time}} \cdot (1 - (x_{ij} + y_{ij})) \\ \forall\, i, j \in I,\ i \neq j \end{gathered} \tag{9}$$

$$u_{10} \leq T^{max} \tag{10}$$

$$a_i \leq u_i \leq b_i \quad \forall i \in V \tag{11}$$

$$tw_{ij} > 30 \Rightarrow x_{ij} = 0 \tag{12}$$

5.4.3. Budget Constraints

$$c_0 = 0 \tag{13}$$

$$c_j \geq c_i + (Ticket \cdot y_{ij}) + (Ticket\_M_j \cdot (x_{ij} + y_{ij})) - M_{\text{cost}} \cdot (1 - (x_{ij} + y_{ij})) \\ \forall\, i, j \in I,\ i \neq j \tag{14}$$

$$c_{10} \leq C^{max} \tag{15}$$

5.5. Tourist Attractions

The set of tourist attractions considered in the model consists of the following nine cultural and historical sites in Paris:

Node	Attraction
1	Musée du Louvre
2	Musée d'Orsay
3	Musée de l'Orangerie
4	Tour Eiffel
5	Arc de Triomphe
6	Sacré-Cœur
7	Opéra Garnier
8	Notre-Dame de Paris
9	Musée Rodin

Table 3: Tourist Attractions Considered in the Analysis

These attractions were selected due to their cultural relevance, historical importance, and high attractiveness for international tourists visiting Paris. The hotel is defined as both the starting and ending point of the itinerary, ensuring a closed-loop route structure.

6. Model Characteristics

- Mixed-Integer Linear Programming (MILP) formulation
- Solvable using CPLEX, Gurobi, or GLPK
- Multimodal transportation structure (walking and metro)
- Temporal propagation constraints eliminate subtours
- Fully linear mathematical structure

7. Distance and Time Assumptions

Walking times are computed using Euclidean distances between coordinates and a calibrated walking speed of 5 km/h (approximately 12 min/km). However, for greater realism and scalability, the proposed framework integrates OpenStreetMap-based routing data to obtain real street-network distances and more accurate travel times between attractions.

This approach allows the model to better capture actual urban mobility conditions, including street connectivity and pedestrian pathways, improving the quality and feasibility of the generated itineraries.

8. Business Value and Strategic Impact

The proposed framework creates value for multiple stakeholders. While the primary users of the algorithm are expected to be tourists, there are positive spillover effects for other actors who may benefit both directly and indirectly from its implementation. These benefits extend beyond individual travelers, contributing to improvements in tourism platforms, urban mobility systems, and local service providers.

8.1. Tourists

- Improved travel satisfaction
- Reduced itinerary planning complexity
- Better time utilization
- Lower transportation inefficiencies
- Better budget utilization

8.2. Tourism Platforms

- Personalized itinerary generation
- Increased platform differentiation
- Higher user engagement

8.3. Smart Cities

- Improved tourist flow distribution
- Better urban mobility management
- Reduced congestion in overcrowded areas

8.4. Travel Agencies and Hospitality Providers

- Automated premium itinerary services
- Data-driven recommendations
- Improved customer experience

9. Implementation Opportunities

The proposed framework presents multiple opportunities for real-world implementation across the tourism, mobility, and smart city sectors.

9.1. Tourism Platforms

The optimization engine could be integrated into existing tourism applications to provide dynamically generated personalized itineraries based on traveler preferences, available time, and budget constraints. These tourism platforms may be either private or public, with the main objective of efficiently handling complex constraints within a single integrated system.

9.2. Mobile Applications

The framework could support mobile itinerary assistants capable of updating routes in real time according to congestion, weather conditions, transportation disruptions, or attraction availability. In addition, users could directly benefit from the algorithm through mobile applications that facilitate trip planning, enhance navigation efficiency, and improve the overall travel experience.

9.3. Smart City Systems

City governments and tourism authorities could use the model to improve tourist flow distribution, reduce overcrowding at highly congested attractions, and promote more balanced urban mobility patterns. In addition, tourism agencies and visitor centers could promote specific destinations based on their characteristics and visitor preferences, helping increase tourist visits to less explored or emerging attractions.

9.4. Hospitality and Travel Services

Hotels, travel agencies, and concierge services could leverage the framework to automatically generate premium itineraries tailored to customer profiles and trip duration, enabling more personalized travel experiences and improving operational efficiency through automated route planning and adaptive scheduling.

9.5. Future AI Integration

Future versions of the framework could incorporate machine learning and artificial intelligence components, including:

- Preference prediction models
- Real-time adaptive optimization
- Reinforcement learning for itinerary refinement
- Personalized recommendation systems

These extensions would enhance both scalability and personalization capabilities.

10. Data

The data used in this study were collected from the official websites of the tourist attractions presented in the previous sections. Information regarding opening and closing hours was used to define the time window constraints of the optimization model. In addition, ticket prices were obtained from the same official sources in order to incorporate budget-related restrictions into the framework. For the geographical location and coordinates of the attractions, free-access data provided by OpenStreetMap were used.

11. Results

From a technical perspective, the optimization model was executed with a time limit of 180 seconds and an MIP gap of 0.15. The time limit was introduced to prevent excessive computational times, since the model may require a long execution period to converge to an optimal solution due to the complexity of the routing and scheduling constraints. Therefore, these parameters allow the solver to obtain a high-quality feasible solution within a reasonable computational effort.

The optimization was performed considering the tourist preferences provided by the user in the form of a ranking of the Points of Interest (POIs). Each attraction was assigned a score on a scale from 1 to 10, where higher values represent greater preference or interest for the tourist.

Point of Interest (POI)	**Score**
Musée du Louvre	8
Notre-Dame de Paris	6
Musée de l'Orangerie	5
Musée d'Orsay	4
Arc de Triomphe	4
Opéra Garnier	3
Tour Eiffel	2
Sacré-Cœur	2
Musée Rodin	1

Table 4: Tourist preference scores assigned to each POI.

Additionally, the optimization process considered the following global constraints:

- Maximum available budget: 100 euros
- Maximum available time: 600 minutes (10 hours)
- Depot: Hôtel du Louvre

Based on these constraints and the preference scores assigned to each attraction, the algorithm generated an optimized tourist route that maximizes the overall utility of the itinerary while satisfying both the budget and time limitations.

The following route represents the solution obtained by the model, and the corresponding visual representation of the optimized route is presented below.

As a consequence of the optimization process, the following attractions were excluded from the final itinerary:

- Tour Eiffel
- Musée Rodin
- Musée d'Orsay

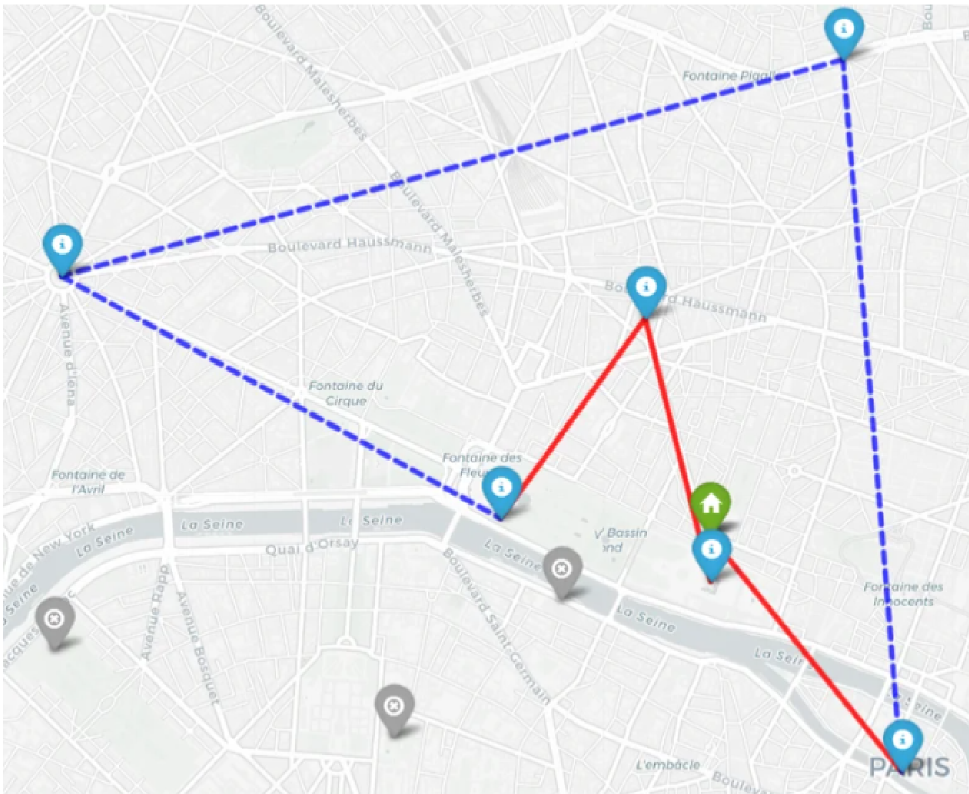

Figure 2: Optimal Tourist Route in Paris

From	**To**	**Transport**	**Arrival** /	**Ticket**
Hotel (Louvre)	Notre-Dame de Paris	By foot (21 min)	08:20	16€
Notre-Dame de Paris	Sacré-Cœur	Metro (15 min)	10:04	8€
Sacré-Cœur	Arc de Triomphe	Metro (22 min)	11:25	16€
Arc de Triomphe	Musée de l'Orangerie	Metro (15 min)	12:25	11€
Musée de l'Orangerie	Opéra Garnier	By foot (16 min)	13:41	15€
Opéra Garnier	Musée du Louvre	By foot (16 min)	14:57	22€
Musée du Louvre	Return Hotel	By foot (3 min)	18:00	—

Table 5: Optimized Tour in Paris *(Budget: 100 €, Total score: 28 pt, Total cost: 95.65 €)*

The optimized tour achieves a total score of 28 points, with a metro cost of 7.65€ and a POI cost of 88.00€, resulting in a total expenditure of 95.65€ within the 100€ budget. The itinerary runs from 08:00 to 18:00.

Some attractions were discarded because their inclusion would reduce the overall efficiency of the route under the imposed constraints and the user-defined preference structure. This outcome is mainly driven by their relatively large distance from the main cluster of selected POIs, as well as the inherent trade-off between travel time and attractiveness scores within the optimization process.

12. Sensitivity analysis

In addition, a sensitivity analysis was performed to evaluate the robustness of the model. The model was tested under different runtime limits (3 minutes, 10 minutes, and 1 hour). The obtained solution remained unchanged across all computational settings, indicating a high degree of stability and robustness of the optimization framework. As a result, the same set of attractions was consistently excluded from the final itinerary.

To further evaluate the robustness and stability of the proposed optimization model, a sensitivity analysis was conducted by increasing the available budget by 185 while keeping all the other parameters unchanged. In particular, different runtime limits were tested in order to assess the impact of computational time on the quality of the generated itinerary and the selection of Points of Interest (POIs).

12.0.1. Runtime Limit: 180 Seconds

For the first analysis, the solver runtime limit was set to 180 seconds, while all the remaining parameters were maintained unchanged.

Under this configuration, the algorithm excluded the same POIs that were omitted in the original solution:

- Eiffel Tower
- Musée Rodin
- Musée d'Orsay

Additionally, the model also excluded:

- Sacré-Cœur

The resulting itinerary finished significantly earlier than the available 10-hour time horizon. This outcome suggests that the algorithm was not able to fully exploit the available visiting time within the imposed computational limit. Consequently, the relatively short runtime restricted the exploration of the feasible solution space, preventing the solver from identifying more complete and higher-quality routes. Therefore, the results indicate that insufficient computational time may negatively affect the efficiency and completeness of the generated itinerary.

The following presents a summary of the optimal solution, including the table and the map.

From	To	Transport	Arrival	Ticket
Hotel (Louvre)	Musée du Louvre	By foot (3 min)	09:00	22 €
Musée du Louvre	Arc de Triomphe	Metro (19 min)	12:19	16 €
Arc de Triomphe	Notre-Dame de Paris	Metro (24 min)	13:28	16 €
Notre-Dame de Paris	Musée de l'Orangerie	Metro (14 min)	14:27	11 €
Musée de l'Orangerie	Opéra Garnier	By foot (16 min)	15:43	15 €
Opéra Garnier	Return Hotel	By foot (13 min)	16:56	–

Table 6: Optimized tourist itinerary in Paris *(Budget: 185 €, Total score: 26 pt, Total cost: 87.65 €).*

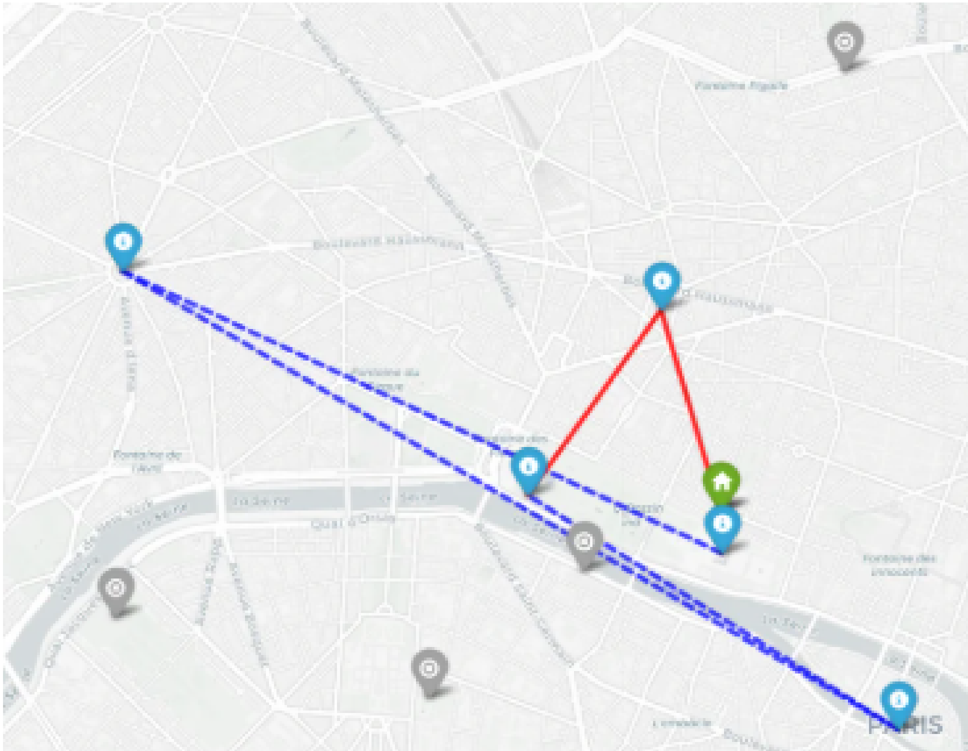

Figure 3: Optimized Paris Tourist Route *(Budget = €185, Runtime = 180s)*

12.0.2. Runtime Limit: 600 Seconds

In the second analysis, the runtime limit was increased to 600 seconds while maintaining all other parameters unchanged.

With the additional computational time, the algorithm generated a different itinerary. In this case, the excluded POIs were:

- Sacré-Cœur
- Musée d'Orsay

- Eiffel Tower

Compared to the previous solution obtained before increasing the budget, Musée Rodin was now included in the itinerary instead of Sacré-Cœur. This result suggests that providing the solver with more runtime improves the quality of the solution, allowing the algorithm to explore a larger number of feasible combinations of POIs while still satisfying all the imposed constraints.

Furthermore, the change in the selected attractions demonstrates that the optimization problem is highly sensitive to computational time. Even moderate increases in runtime can lead to substantially different itineraries and improved resource utilization.

The following presents a summary of the optimal solution, including the table and the map.

From	**To**	**Transport**	**Arrival**	**Ticket**
Hotel (Louvre)	Notre-Dame de Paris	By foot (21 min)	08:20	16 €
Notre-Dame de Paris	Musée Rodin	Metro (16 min)	10:00	15 €
Musée Rodin	Arc de Triomphe	Metro (14 min)	11:14	16 €
Arc de Triomphe	Opéra Garnier	Metro (17 min)	12:16	15 €
Opéra Garnier	Musée de l'Orangerie	By foot (16 min)	13:33	11 €
Musée de l'Orangerie	Musée du Louvre	By foot (18 min)	14:50	22 €
Musée du Louvre	Return Hotel	By foot (3 min)	18:00	–

Table 7: Optimized tourist itinerary in Paris *(Budget: 185 €, Total score: 27 pt, Total cost: 185 €).*

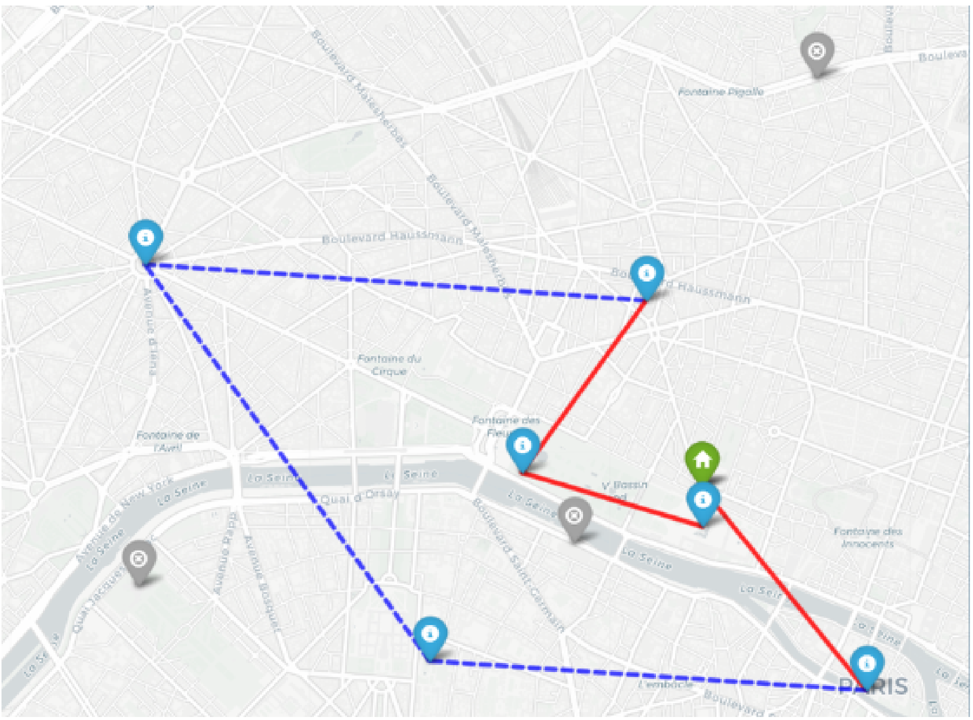

Figure 4: Optimized Paris Tourist Route *(Budget = €185, Runtime = 600s)*

12.0.3. Runtime Limit: 3600 Seconds

Finally, the runtime limit was extended to 3600 seconds in order to analyze the model's behavior under a significantly larger computational budget.

Under this configuration, the solution changed once again. In particular, Sacré-Cœur was included in the itinerary, while Musée Rodin was excluded.

This result highlights the strong influence of runtime on the final solution obtained by the optimization model. As the computational time increases, the algorithm is able to explore a broader portion of the feasible solution space and evaluate additional alternative combinations of POIs. Consequently, the generated itinerary becomes more refined and potentially closer to the global optimum.

The following presents a summary of the optimal solution, including the table and the map.

From	**To**	**Transport**	**Arrival**	**Ticket**
Hotel (Louvre)	Notre-Dame de Paris	By foot (21 min)	08:20	16 €
Notre-Dame de Paris	Musée de l'Orangerie	Metro (14 min)	09:19	11 €
Musée de l'Orangerie	Opéra Garnier	By foot (16 min)	10:36	15 €
Opéra Garnier	Arc de Triomphe	Metro (17 min)	11:53	16 €
Arc de Triomphe	Sacré-Cœur	Metro (22 min)	13:44	8 €
Sacré-Cœur	Musée du Louvre	Metro (13 min)	14:57	22 €
Musée du Louvre	Return Hotel	By foot (3 min)	18:00	–

Table 8: Optimized tourist itinerary in Paris *(Budget: 185 €, Total score: 28 pt, Total cost: 98.20 €).*

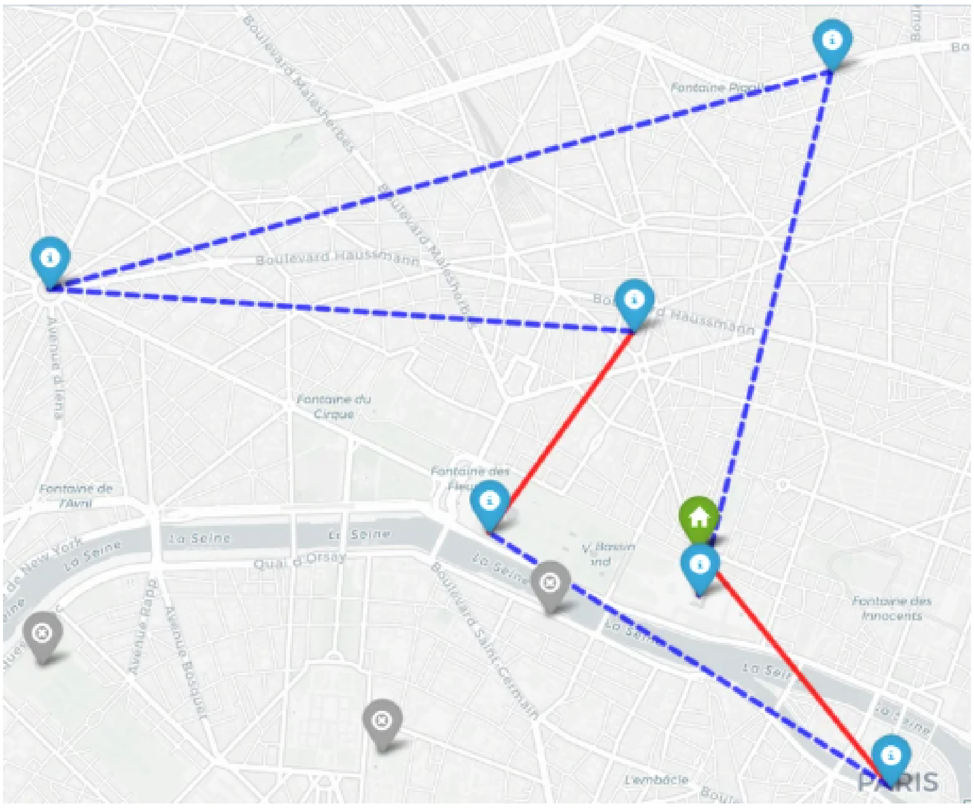

Figure 5: Optimized Paris Tourist Route *(Budget = €185, Runtime = 3600s)*

Overall, the sensitivity analysis demonstrates that the runtime limit has a substantial impact on the performance and effectiveness of the optimization process. Increasing the available computational time allows the solver to identify more diversified and higher-quality solutions, improving the balance between budget utilization, visiting time, and attraction selection. Therefore, the analysis confirms the importance of carefully balancing computational resources and solution quality when solving complex itinerary optimization problems.

		Budget: 100	**Budget: 185**		
Attraction	**Score**	**180 s**	**180 s**	**600 s**	**3 600 s**
Musée du Louvre	8	✓	✓	✓	✓
Notre-Dame de Paris	6	✓	✓	✓	✓
Musée de l'Orangerie	5	✓	✓	✓	✓
Arc de Triomphe	4	✓	✓	✓	✓
Opéra Garnier	3	✓	✓	✓	✓
Sacré-Cœur	2	✓	×	×	✓
Tour Eiffel	2	×	×	×	×
Musée d'Orsay	4	×	×	×	×
Musée Rodin	1	×	×	✓	×
Total score		**28**	**26**	**27**	**28**

Table 9: Included attractions by budget and runtime limit. ✓ = included, × = excluded.

The results presented in Table 6 reveal an important relationship between budget size and the computational resources required to obtain a high-quality solution. Under the original budget of 100 €, the solver reaches a total score of 28 points within the 180-second runtime limit, the model was tested under different runtime limits (3 minutes, 10 minutes, and 1 hour), and the obtained solution remained unchanged across all computational settings, indicating a high degree of stability under the original budget. However, when the budget is increased to 185 €, the feasible solution space expands considerably, as a larger number of POI combinations and routing alternatives become financially viable. This added complexity makes the optimization problem harder to solve, and the solver struggles to exploit the additional budget effectively under tight time constraints: at 180 seconds it achieves only 26 points, and at 600 seconds only 27 points, both below the baseline result. Only when the runtime limit is extended to 3 600 seconds does the solver recover a solution of equivalent quality (28 points). It is worth noting that 28 points does not necessarily represent the true optimum for the 185 € budget. Since a larger budget can never theoretically worsen the optimal objective value, a higher-quality solution likely exists but has not yet been identified within the tested runtime limits. This is consistent with the MIP gap of 0.15 used throughout the analysis, which allows the solver to terminate before proving global optimality. Taken together, these findings underscore a key insight of the sensitivity analysis: increasing the available budget does not automatically translate into a better itinerary unless it is accompanied by a proportional increase in computational time. Careful calibration of both parameters is therefore essential when applying this optimization framework to practical tourist routing problems. It is important to note that, although the same level of satisfaction (28 points) is achieved under both budget scenarios, there is a small but non-negligible difference in expenditure (approximately €2.55 higher under the €185 budget), as well as a change in the optimal visitation order of the selected attractions, reflecting that multiple near-optimal routing configurations exist even when the final objective value remains unchanged.

13. Sustainability and Urban Mobility Impact

Beyond tourism optimization, the proposed framework contributes to more sustainable urban mobility practices.

By prioritizing walking whenever operationally feasible and minimizing unnecessary transportation usage, the model promotes lower-emission mobility patterns within dense urban environments.

The framework also contributes to:

- Reduced transportation congestion
- More balanced tourist flow distribution
- Improved pedestrian accessibility
- More efficient utilization of public transportation systems

From a smart city perspective, optimization-driven tourism planning may help urban authorities better manage overcrowding, improve visitor experience, and reduce pressure on highly concentrated tourism zones.

The integration of sustainability objectives into future versions of the model could further support environmentally responsible tourism strategies.

14. Conclusion

This project demonstrates how mathematical optimization techniques can improve urban tourism itinerary planning under realistic operational constraints. The proposed MILP-based framework successfully generated a feasible and satisfaction-maximizing itinerary for Paris, visiting three attractions, achieving a total preference score of 28 points, and utilizing 95.65€ of the 100€ budget within the available time window.

By integrating user preferences, budget limitations, attraction schedules, and multimodal transportation decisions into a unified framework, the model produces operationally viable itineraries that outperform traditional heuristic planning approaches. The sensitivity analysis further confirmed the robustness of the solution, as results remained stable across different computational time limits, suggesting that the model converges reliably under the imposed constraints.

Despite its promising performance, the framework operates under simplifying assumptions, including Euclidean distance approximations and static operating schedules, which may affect the accuracy of results under real urban conditions. These limitations, however, represent natural directions for future development rather than fundamental weaknesses of the proposed approach.

From a broader perspective, the MILP formulation is not specific to Paris and can be directly transferred to other metropolitan tourism environments with minimal adaptation. Future extensions incorporating real-time data, machine learning-based preference prediction, and adaptive re-optimization could significantly enhance both the scalability and personalization capabilities of the framework, positioning it as a practical tool for smart tourism and urban mobility applications.

15. Limitations and Future improvements

Despite the promising results obtained from the optimization model, several limitations should be acknowledged. Due to computational constraints, the dataset includes only a subset of the available Points of Interest (POIs) in Paris, which may reduce the overall representativeness of the solution. In addition, distances between locations are currently calculated using Euclidean metrics rather than real pedestrian routes, leading to an approximation of actual travel conditions. Another limitation concerns opening and closing times, as operating hours may vary depending on the month and specific closing days are not yet considered in the model. For example, the Louvre Museum is closed every Tuesday. Therefore, if a tourist assigns a high score to the Louvre on a Tuesday in order to obtain an optimized tour, the model may still include it in the final solution even though it cannot actually be visited on that day. For this reason, tourists should carefully verify the itinerary and the opening schedules of the selected POIs even after obtaining the optimized route.

Finally, considering that this project was originally conceived as the core engine for a consumer mobile application, computational execution speed is a critical factor since it directly impacts the user experience. In a real-world context, considering a heuristic resolution framework would be highly preferable, due to the fact that it allows the system to retrieve a high quality sub-optimal solution in a very small span of time. For the scope of this study, an exact solver was maintained primarily to guarantee operational benchmark clarity and to analyse the true mathematical global optimum of the route under a limited dataset. However, as the database scales to incorporate a larger and more realistic number of POIs across Paris, the exact model encounters exponential computational complexity. In such dense network scenarios, implementing a heuristic or metaheuristic algorithm becomes inevitable to provide an immediate and valuable trip plan for the user, but also to avoid huge waiting times for the outputs. For the scope of this study, an exact solver was maintained primarily to guarantee operational benchmark clarity and to analyse the true mathematical global optimum of the route under a limited dataset. However, as the database grows to incorporate a larger and more realistic number of POIs across Paris, the combinations of binary decision variables for transport modes and sequencing face an exponential explosion due to the nature of the problem. In such dense network scenarios, where real-time responsiveness is mandatory, implementing a heuristic approach becomes both a strategic advantage for user retention and a mathematical necessity to bypass the computational bottleneck of exact algorithms.

Future improvements could enhance the realism and accuracy of the results by adjusting travel dynamics, for example by reducing the average walking speed to better reflect realistic pedestrian movement. Furthermore, introducing variability in travel times would allow the model to better capture real-life uncertainties such as traffic conditions, crowd density, and unforeseen delays.